Abstract

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Chapter 1

Exponential Sums over Finite Fields

Exponential sums are important tools in number theory for solving problems involving integers. These sums have many applications in Analytic Number Theory.

In this chapter we will set up exponential sums for finite fields by special group homomorphisms called *characters*. Since a finite field has two different inherent group structure; one is additive and the other is multiplicative we obtain two different characters. Exponential sums are formed by using the values of one or more characters and possibly combining them with weights or with other function values.

1.1 Characters of Finite Abelian Groups

Let G be a finite abelian group with the identity element 1_G. Let S^1 be the set of complex numbers with absolute value 1. Then consider the multiplicative group S^1 with the operation of multiplication.

Definition 1.1.1: Character χ of G

A character χ *of* G *is a group homomorphism* $\chi : G \longrightarrow S^1$ *to the multiplicative group* S^1 *i.e. for all* $g_1, g_2 \in G$

$$\chi(g_1 \cdot g_2) = \chi(g_1) \cdot \chi(g_2)$$

So if χ is any character of G then $\chi(1_G) = 1$. Since by definition for all $g \in G$,

$$\chi(g)^{|G|} = \chi(g^{|G|}) = \chi(1_G) = 1$$

the values of χ are $|G|^{th}$ roots of unity. Also we have for any $g \in G$,

$$\chi(g)\chi(g^{-1}) = \chi(g \cdot g^{-1}) = \chi(1_G) = 1$$

so $\chi(g^{-1}) = \chi(g)^{-1} = \overline{\chi(g)}$.

There are some special characters of G which have specific name. The character χ_0 defined by $\chi_0(g) = 1$ for all $g \in G$ is called the *trivial character*. For any character χ of G, the character $\overline{\chi}$ defined by $\overline{\chi}(g) = \overline{\chi(g)}$ for all $g \in G$ is called the *conjugate character* of χ.

Given any finitely many characters of G, $\chi_1, \ldots, \chi_n$ we can define the product character $\chi = \chi_1 \cdot \chi_2 \cdots \chi_n$ such that for all $g \in G$,

$$\chi(g) = \chi_1(g) \cdot \chi_2(g) \cdots \chi_n(g)$$

Hence for all $g \in G$,

$$\chi(g) \cdot \overline{\chi}(g) = |\chi(g)|^2 = 1$$

Hence the $\overline{\chi}$ is also the inverse character of χ. Hence the set of all characters of G forms a group, $\mathrm{Hom}(G, S^1)$ under multiplication whose identity element is the trivial character χ_0.

Definition 1.1.2: Dual Group to G

For a finite abelian group G, the group of all characters of G with the identity element χ_0 is called the dual group *to G and is denoted by $\widehat{G}$ i.e. $\widehat{G} := \mathrm{Hom}(G, S^1)$.*

Observation 1.1. *The dual group $\widehat{G}$ is a finite abelian group. So we can talk about characters of $\widehat{G}$ too.*

Since the values of characters of G can only be $|G|^{th}$ roots of unity, the dual group $\widehat{G}$ is a finite abelian group.

Lemma 1.1.1

If G is a finite cyclic group of order n and let g be the generator of G, then for all $j \in \{0, 1, \ldots, n-1\}$ define the function

$$\chi_j(g^k) = e(j \cdot k/n) \quad \forall\, k \in \{0, 1, \ldots, n-1\}$$

Then $\widehat{G} = \{\chi_j : j \in \{0, 1, \ldots, n-1\}$.

Proof: Certainly for all $j \in \{0, 1, \ldots, n-1\}$ defines a character of G. On the other hand suppose, $\chi \in \widehat{G}$. Then $\chi(g)$ must be an nth root of unity. Hence there exists $j \in \{0, 1, \ldots, n-1\}$ such that $\chi(g) = e(j/n)$. Therefore $\chi = \chi_j$. Hence $\widehat{G} = \{\chi_j : 0 \leq j \leq n-1\}$. ■

Observation 1.2. *The characters defined in above lemma forms a cyclic group with χ_1 being the generator of $\widehat{G}$.*

Corollary 1.1.2

If G is a finite cyclic group of order n with g being the generator then $G \cong \widehat{G}$ by the isomorphism $\varphi : g^j \mapsto \chi_j$ for all $j \in \{0, 1, \ldots, n-1\}$.

Now we will show that we can extend a character of any subgroup to a character of the group.

Theorem 1.1.3

Let H be a subgroup of the finite abelian group G and let χ_H be a character of H. Then χ_H can be extended to a character of G that is there exists a character $\chi \in \widehat{G}$ with $\chi(h) = \chi_H(h)$ for all $h \in H$.

Proof: Let H be a proper subgroup of G. Let $\alpha \in G \setminus H$. Then let H_1 be the group generated by H and α i.e. $H_1 = \langle H, a \rangle$. Let m be the smallest positive integer such that $\alpha^m \in H$. Then for all $g \in H_1$, g can be written uniquely as $g = \alpha^j \cdot h$ where $0 \leq j < m$ and $h \in H$.

Let ω be a complex number such that $\omega^m = \chi_H(\alpha^m)$. Then for all $g \in H_1$ where $g = \alpha^j \cdot h$, $h \in H$ define the function $\chi_1(g) = \omega^j \chi_H(h)$. Then for any $g_1, g_2 \in H_1$ where $g_1 = \alpha^i \cdot h_1$, $g_2 = \alpha^j \cdot h_2$, we have $g_1 \cdot g_2 = \alpha^{j+i} \cdot h_1 \cdot h_2 = \alpha^{j+k \bmod m} \cdot h_1 \cdot h_2$. Hence

$$\chi_1(g_1 \cdot \chi_2) = \omega^{j+k \bmod m} \chi_H(h_1) \cdot \chi_H(h_2) = \alpha^j \chi_H(h_1) \cdot \omega^k \chi_H(h_2) = \chi_1(g_1) \cdot \chi_1(g_2)$$

Hence χ_1 is a character of H_1. And also for all $h \in H$ we have $\chi_1(h) = \chi_H(h)$.

So if $H_1 = G$ we are done. Otherwise we can continue this process above until we get G. This process we have to do finitely many times as G is a finite group. ■

Therefore if we have two distinct elements $g_1, g_2 \in G$, $g_1 \neq g_2$ then take the subgroup generated by the element $h = g_1 \cdot g_2^{-1} \neq 1$. Since this subgroup is a cyclic subgroup generated by h, by Lemma 1.1.1 we get a character χ' of the cyclic subgroup such that $\chi'(h) \neq 1$. Then by Theorem 1.1.3 we get a character χ of G such that $\chi(h) = \chi'(h) \neq 1$ and hence $\chi(g_1) \neq \chi(g_2)$. Therefore we have the following observation

Observation 1.3. *For any two distinct elements $g_1, g_2 \in G$, $g_1 \neq g_2$ we can get a character $\chi \in \widehat{G}$ such that $\chi(g_1) \neq \chi(g_2)$.*

Now consider any character χ of G. If $g \in G$ such that $\chi(g) = 1$ then $\chi(g^{-1}) = 1$. Hence the set of all elements $g \in G$ such that $\chi(g)$ forms a group. This group is called the *annihilator* of χ and is denoted by $\mathrm{Ann}(\chi)$.

Similarly suppose H is a subgroup of G. Then if $\chi \in \widehat{G}$ is a character of G such that $\chi(h) = 1$ for all $h \in H$ then $\overline{\chi}(h) = 1$ for all $h \in H$. Hence the set of all characters $\chi \in \widehat{G}$ such that $\chi(h) = 1$ for all $h \in H$ forms a group. This group is called the *annihilator* of H and is denoted by $\mathrm{Ann}(H)$. Since in both cases the notation Ann is used usually the context will make it clear whether we are talking about the annihilator of a character or the annihilator of a subgroup.

Definition 1.1.3: Annihilator

Let G be a finite abelian group. For any character $\chi \in \widehat{G}$, the annihilator *of χ, $\mathrm{Ann}(\chi) = \{g \in G : \chi(g) = 1\}$ which is a subgroup of G.*

For any subgroup H of G, the annihilator *of H, $\mathrm{Ann}(H) = \{\chi \in \widehat{G} : \chi(h) = 1 \text{ for all } h \in H\}$ which is a subgroup of $\widehat{G}$.*

Now every element of $\widehat{G}$ is a function χ which takes an element of G and maps to a complex number in S^1. Similarly we can think of every element of as a function from $\widehat{G}$ to S^1. In fact for every element $g \in G$ define the homomorphism $\mathrm{ev}_g : \widehat{G} \to S^1$ where $\mathrm{ev}_g(\chi) = \chi(g)$. Hence ev_g is a character of $\widehat{G}$ for all $g \in G$.

Theorem 1.1.4

Let G be a finite abelian group.

(i) If $\chi \in \widehat{G}$ be any nontrivial character then

$$\sum_{g \in G} \chi(g) = 0 \tag{1.1}$$

(ii) If $g \in G$ with $g \neq 1_G$ then

$$\sum_{\chi \in \widehat{G}} \chi(g) = 0 \tag{1.2}$$

Proof: Since χ is nontrivial there exists $h \in G$ such that $\chi(h) \neq 1$. Then

$$\chi(h) \sum_{g \in G} \chi(g) = \sum_{g \in G} \chi(h \cdot g) = \sum_{g \in G} \chi(g)$$

Hence we have

$$(\chi(h) - 1) \sum_{g \in G} \chi(g) = 0$$

Since we know $\chi(h) \neq 1$ we have $\sum_{g \in G} \chi(g) = 0$.

Now from the discussion above for any $g \in G$ and $\chi \in hG$ we have $\chi(g) = \mathrm{ev}_g(\chi)$. Hence

$$\sum_{\chi \in \widehat{G}} \chi(g) = \sum_{\chi \in \widehat{G}} \mathrm{ev}_g(\chi)$$

Since $g \neq 1_G$, there is a $\chi \in \widehat{G}$ such that $\chi(g) \neq 1$. Hence ev_g is a nontrivial character of $\widehat{G}$. Hence by the first part of the theorem we have $\sum_{\chi \in \widehat{G}} \mathrm{ev}_g(\chi) = 0$. ∎

The above theorem leaves the case when χ is the trivial character. Then for all $g \in G$, $\chi(g) = 1$. Hence we have $\sum_{g\in G} \chi(g) = |G|$. Hence we have the following corollary

Observation 1.4. *For any finite abelian group G. Then*

(i) Let χ is a character of G.

$$\sum_{g\in G} \chi(g) = \begin{cases} |G| & \text{if } \chi = \chi_0 \\ 0 & \text{otherwise} \end{cases} \tag{1.3}$$

(ii) $|G| = \sum_{\chi\in\widehat{G}} \sum_{g\in G} \chi(g)$

So now by counting in two ways for the sum $\sum_{\chi\in\widehat{G}} \sum_{g\in G} \chi(g)$ we get the following theorem:

Theorem 1.1.5

For any finite abelian group G, we have $|G| = |\widehat{G}|$.

Proof: From Observation 1.4(ii) we have

$$|G| = \sum_{\chi\in\widehat{G}} \sum_{g\in G} \chi(g) = \sum_{g\in G} \sum_{\chi\in\widehat{G}} \chi(g) = |\widehat{G}|$$

Hence we have the result. ■

From above theorem and Observation 1.4(i) we get the *orthogonality relations for characters.*

Observation 1.5. *Let G be a finite abelian group.*

(i) Let χ, τ are characters of G. Then

$$\frac{1}{|G|} \sum_{g\in G} \chi(g) \cdot \overline{\tau(g)} = \begin{cases} 1 & \text{if } \chi = \tau \\ 0 & \text{otherwise} \end{cases}$$

(ii) If $g, h \in G$ are any two elements then

$$\frac{1}{|G|} \sum_{\chi\in\widehat{G}} \chi(g) \cdot \overline{\chi(h)} = \begin{cases} 1 & \text{if } g = h \\ 0 & \text{otherwise} \end{cases}$$

Since we already showed that for every element $g \in G$, ev_g is a character of $\widehat{G}$. Theorem 1.1.5 gives us the possibility that G and $\widehat{G}$ are isomorphic. In Lemma 1.1.1 we already showed that for cyclic groups G and $\widehat{G}$ are isomorphic. In the next few results we will show that this is the case for all finite abelian groups.

By Fundamental Theory of Finite Abelian Groups every finite abelian group G can be written as a direct product of cyclic groups. Hence it is enough to show that if G_1 and G_2 are two finite abelian groups and $G = G_1 \times G_2$ then $\widehat{G}$ is isomorphic to the direct product of $\widehat{G}_1$ and $\widehat{G}_2$.

Theorem 1.1.6

Let G_1 and G_2 be two finite abelian groups and let $G = G_1 \times G_2$. Then $\widehat{G} \cong \widehat{G}_1 \times \widehat{G}_2$.

Proof: Since G is the direct product of G_1 and G_2, G consists of pairs (g_1, g_2) where $g_1 \in G_1$ and $g_2 \in G_2$. So consider the map $\varphi : \widehat{G}_1 \times \widehat{G}_2 \to \widehat{G}$ where for any $\chi_1 \in \widehat{G}_1$ and $\chi_2 \in \widehat{G}_2$, $\chi := \varphi(\chi_1, \chi_2)$ is the character of G defined by the following:

$$\text{for all } (g_1, g_2) \in G \qquad \chi(g_1, g_2) = \chi(g_1, 1) \cdot \chi(1, g_2) = \chi_1(g_1) \cdot \chi_2(g_2)$$

Hence φ is a homomorphism. Now we will show that φ is an isomorphism.

Let $\chi \in \widehat{G}$. Then define for all $\chi_1(g_1) = \chi(g_1, 1)$ for all $g_1 \in G_1$ and similarly define $\chi_(g_2) = \chi(1, g_2)$ for all $g_2 \in G_2$. Then χ_1 is a character of G_1 and χ_2 is a character of G_2. And also for any $(g_1, g_2) \in G$ we have

$$\chi_1(g_1) \cdot \chi_2(g_2) = \chi(g_1, 1) \cdot \chi(1, g_2) = \chi(g_1, g_2)$$

Hence $\varphi(\chi_1, \chi_2) = \chi$. Hence φ is an isomorphism. Therefore $\widehat{G} \cong \widehat{G}_1 \times \widehat{G}_2$. ■

Theorem 1.1.7

For any finite abelian group G, G is isomorphic to $\widehat{G}$.

Proof: By Fundamental Theorem of Finite Abelian Groups, G can be written as a direct product of cyclic groups i.e. there exists cyclic groups $Z_{n_1}, Z_{n_2}, \ldots, Z_{n_k}$ such that

$$G \cong Z_{n_1} \times Z_{n_2} \times \cdots \times Z_{n_k}$$

Hence by repeated use of Theorem 1.1.6 we have $\widehat{G} \cong \widehat{Z}_{n_1} \times \widehat{Z}_{n_2} \times \cdots \times \widehat{Z}_{n_k}$. Since by Lemma 1.1.1 for all $i \in \{1, 2, \ldots, k\}$, $\widehat{Z}_{n_i} \cong Z_{n_i}$ we have $\widehat{G} \cong Z_{n_1} \times Z_{n_2} \times \cdots \times Z_{n_k} \cong G$ Hence we have the result. ■

1.2 Characters of Finite Fields

In a finite field $\mathbb{F}_q$ there are two inherent group structures–namely, the additive group and the multiplicatve group. The nonzero elements of $\mathbb{F}_q$ forms a cyclic group of $q-1$ elements under multiplication and the whole $\mathbb{F}_q$ forms a group of q elements under addition.

1.2.1 Additive Characters of $\mathbb{F}_q$

We first turn to the additive group of $\mathbb{F}_q$. Characters of the additive group $\mathbb{F}_q$ are called *additive characters* of $\mathbb{F}_q$. We denote the set of all additive characters of $\mathbb{F}_q$ by $\boldsymbol{\mathcal{X}}_q$. Let p be the characteristic of $\mathbb{F}_q$. Then the prime field contained in $\mathbb{F}_q$ is $\mathbb{F}_p$.

Let $\mathrm{Tr} : \mathbb{F}_q \to \mathbb{F}_p$ be the absolute trace function from $\mathbb{F}_q$ to $\mathbb{F}_p$. Then define the function χ_1 by

$$\chi_1(\alpha) = e^{2\pi i \frac{\mathrm{Tr}(\alpha)}{p}} \quad \forall\, \alpha \in \mathbb{F}_q$$

Then χ_1 is an additive character of $\mathbb{F}_q$ since for all $\alpha_1, \alpha_2 \in \mathbb{F}_q$, $\mathrm{Tr}(\alpha_1 + \alpha_2) = \mathrm{Tr}(\alpha_1) + \mathrm{Tr}(\alpha_2)$ and hence $\chi_1(\alpha_1 + \alpha_2) = \chi_1(\alpha_1) \cdot \chi_1(\alpha_2)$. This character χ_1 is called the *canonical additive character* of $\mathbb{F}_q$.

Theorem 1.2.1

For $\beta \in \mathbb{F}_q$, define the function χ_β with $\chi_\beta(\alpha) = \chi_1(\beta \cdot \alpha)$ for all $\alpha \in \mathbb{F}_q$. Then χ_β is an additive character of $\mathbb{F}_q$ and every additive character of $\mathbb{F}_q$ is obtained this way.

Proof: Now for any $\alpha_1, \alpha_2 \in \mathbb{F}_q$, we have

$$\chi_\beta(\alpha_1 + \alpha_2) = \chi_1(\beta(\alpha_1 + \alpha_2)) = \chi_1(\beta \cdot \alpha_1) \cdot \chi_1(\beta \cdot \alpha_2) = \chi_\beta(\alpha_1) \cdot \chi_\beta(\alpha_2)$$

Hence χ_β is an additive character of $\mathbb{F}_q$.

Now we will show that for any two distinct $\beta_1, \beta_2 \in \mathbb{F}_q$ with $\beta_1 \neq \beta_2$ there exists an $\alpha \in \mathbb{F}_q$ such that $\chi_{\beta_1}(\alpha) \neq \chi_{\beta_2}(\alpha)$. Suppose that is not the case. Then for all $\alpha \in \mathbb{F}_q$,

$$\chi_{\beta_2}(\alpha) = \chi_{\beta_2}(\alpha) \iff \chi_1(\beta_1 \cdot \alpha) = \chi_1(\beta_2 \cdot \alpha) \iff \chi_1((\beta_1 - \beta_2)\alpha) = 1$$

. Now $\chi_1((\beta_1 - \beta_2)\alpha) = 1$ for all $\alpha \in \mathbb{F}_q$ if and only if $\mathrm{Tr}((\beta_1 - \beta_2)\alpha) = 0$ for all $\alpha \in \mathbb{F}_q$. Since $\mathrm{Tr}(X) = X + X^p + \cdots + X^{pr-1}$ where $q = p^r$, $\mathrm{Tr}((\beta_1 - \beta_2)X)$ has p^r roots in $\mathbb{F}_q$ which is more than the degree p^{r-1}. Hence $\mathrm{Tr}((\beta_1 - \beta_2)X)$ is the zero polynomial. Therefore $\beta_1 = \beta_2$. Hence for any two distinct $\beta_1, \beta_2 \in \mathbb{F}_q$ with $\beta_1 \neq \beta_2$ there exists an $\alpha \in \mathbb{F}_q$ such that $\chi_{\beta_1}(\alpha) \neq \chi_{\beta_2}(\alpha)$.

Hence for every $\beta \in \mathbb{F}_q$ we have a distinct additive character. Therefore we have q additive characters. Since there are q additive characters of $\mathbb{F}_q$ by Theorem 1.1.5 every additive character of $\mathbb{F}_q$ can be obtained the above way. ■

The orthogonality relations in Observation 1.5 when applied to additive characters and along with Theorem 1.2.1 gives the following result.

Theorem 1.2.2

(i) For any $a \in \mathbb{F}_q$, the corresponding additive character is χ_a. Then

$$\sum_{\gamma \in \mathbb{F}_q} \chi_a(\gamma) = \begin{cases} 0 & \text{if } a \neq 0 \\ q & \text{if } a = 0 \end{cases}$$

(ii) For any two additive characters χ_a, χ_b where $a, b \in \mathbb{F}_q$,

$$\sum_{\gamma \in \mathbb{F}_q} \chi_a(\gamma) \cdot \overline{\chi_b}(\gamma) = \begin{cases} 0 & \text{if } a \neq b \\ q & \text{if } a = b \end{cases}$$

(iii) For any two elements $\alpha, \beta \in \mathbb{F}_q$,

$$\sum_{c \in \mathbb{F}_q} \chi_c(\alpha) \cdot \overline{\chi_c(\beta)} = \begin{cases} 0 & \text{if } \alpha \neq \beta \\ q & \text{if } \alpha = \beta \end{cases}$$

1.2.2 Multiplicative Characters of $\mathbb{F}_q$

Characters of the multiplicate group $\mathbb{F}_q^*$ are called *multiplicative characters* of $\mathbb{F}_q$. We denote the set of all multiplicative characters of $\mathbb{F}_q$ by $\boldsymbol{\mathcal{M}}_q$. Since $\mathbb{F}_q^*$ is a cyclic group of order $q-1$ by Lemma 1.1.1 we have the following:

Lemma 1.2.3

The group of multiplicative characters of $\mathbb{F}_q$ is a cycic group of order $q-1$ and are given by $\{\psi_j : 0 \leq j \leq q-2\}$ where if g is the generator of $\mathbb{F}_q^*$ then for all $k \in \{0, 1, \ldots, q-2\}$ we have $\psi_j(g^k) = e^{2\pi i \frac{jk}{q-1}}$ where ψ_0 is the identity element of the group and the trivial character.

Every multiplicative character ψ of $\mathbb{F}_q$ has the property that $\psi^{q-1} = \psi_0$. Let ψ is generator of the cyclic group of multiplicative characters. Then ψ is called the *principle character*. For any multiplicative character ψ of $\mathbb{F}_q$ we say ψ is of *order d* if $\psi^d = \psi_0$ and d is the smallest positive integer with this property. We use $\mathrm{ord}(\psi)$ to denote the order of ψ. Hence $d \mid q-1$. We say ψ is of *exponent e* is $\psi^e = \psi_0$. Clearly $d \mid e$ where d is the order of ψ. Let $\boldsymbol{\mathcal{M}}_q^{(e)}$ denote the set of all multiplicative characters of $\boldsymbol{\mathcal{M}}_q$ of exponent e.

Remark: For any multiplicative character ψ of $\mathbb{F}_q$, the value of $\psi(-1)$ is of special interest. We obviously have $\psi(-1) = \pm 1$. Let $\operatorname{ord}(\psi) = m$. Then $m \mid q-1$. Hence values of ψ are m^{th} roots of unity. Therefore $\psi(-1)$ can be -1 if m is even. Let g be a primitive element of $\mathbb{F}_q$ i.e. $\mathbb{F}_q^*$ is generated by g. Hence $\psi(g) = \zeta_m$ where ζ_m is the mth root of unity. If m is even

$$\psi(-1) = \psi\left(g^{(q-1)/2}\right) = \zeta_m^{(q-1)/2}$$

Now

$$\zeta_m^{(q-1)/2} = -1 \iff \frac{q-1}{2} \equiv \frac{m}{2} \bmod m \iff \frac{q-1}{m} \equiv 1 \bmod 2$$

So m should be even and $\frac{q-1}{m}$ should be odd. Therefore we have the following observation.

Observation 1.6. *For any multiplicative character $\psi \in \mathcal{M}_q$, $\psi(-1) = -1$ if and only if $\frac{q-1}{m}$ is odd and m is even.*

For odd q a special multiplicative character of interest is the quadratic character. It is denoted by η. It only takes the value ± 1. So $\operatorname{ord}\eta = 2$. For all $\alpha \in \mathbb{F}_q^*$, $\eta(\alpha) = 1$ if α is a quadratic residue and -1 if α is a non-quadratic residue. The *Legendre* symbol $\left(\frac{\alpha}{q}\right)$ for any $\alpha \in \mathbb{F}_q$ is defined by $\eta(\alpha) = \left(\frac{\alpha}{q}\right)$

Suppose $d \mid q-1$. Then for every multiplicative character ψ of exponent d and for every $\alpha \in \mathbb{F}_q^*$, we have $\psi(\alpha^d) = \psi^d(\alpha) = 1$. Hence $\psi(\alpha) = 1$ if $\alpha \in (\mathbb{F}_q^*)^{(d)}$. Conversely if $\psi(\alpha) = 1$ for every $\alpha \in (\mathbb{F}_q^*)^{(d)}$ then $\psi^d = \psi_0$. Hence ψ is a character of exponent d then $\psi(\alpha)$ depends only on the cosets of $\mathbb{F}_q^* / (\mathbb{F}_q^*)^{(d)}$. Therefore there are precisely d such characters of exponent d.

Theorem 1.2.4

Suppose $d \mid q-1$. Then

$$\sum_{\psi \in \mathcal{M}_q^{(d)}} \chi(\alpha) = \begin{cases} d & \text{if } \alpha \in (\mathbb{F}_q^*)^{(d)} \\ 0 & \text{otherwise} \end{cases}$$

Proof: The characters of exponent d are characters of the group $\mathbb{F}_q^* / (\mathbb{F}_q^*)^{(d)}$. Hence this follows from Observation 1.4. ■

Again, the orthogonality relations in Observation 1.5 when applied to multiplicative characters gives the following result.

Theorem 1.2.5

(i) For any multiplicative character ψ of $\mathbb{F}_q$

$$\sum_{\alpha \in \mathbb{F}_q^*} \psi(\alpha) = \begin{cases} 0 & \text{if } \psi \neq \psi_0 \\ q-1 & \text{if } \psi = \psi_0 \end{cases}$$

(ii) For any two multiplicative characters ψ and λ of $\mathbb{F}_q$,

$$\sum_{\alpha \in \mathbb{F}_q^*} \psi(\alpha) \cdot \overline{\lambda(\alpha)} = \begin{cases} 0 & \text{if } \psi \neq \lambda \\ q-1 & \text{if } \psi = \lambda \end{cases}$$

(iii) For any two elements $\alpha, \beta \in \mathbb{F}_q^*$,

$$\sum_{\psi \in \mathcal{M}_q} \psi(\alpha) \cdot \overline{\psi(\beta)} = \begin{cases} 0 & \text{if } \alpha \neq \beta \\ q-1 & \text{if } \alpha = \beta \end{cases}$$

Usually we extend the definition of multiplicative characters to the whole $\mathbb{F}_q$ so that we can define special character sums over the whole field. Let ψ is a multiplicative character then $\psi(0)$ is defined to be 0 if ψ is nontrivial multiplicative character and $\psi(0) = 1$ if ψ is the trivial character.

1.2.3 Lifting of Characters to Finite Extension

Let E be a finite extension field of $\mathbb{F}_q$. Let χ_1 and μ_1 be the canonical additive character of $\mathbb{F}_q$ and E respectively. Let $\mathrm{Tr}_{E/\mathbb{F}_q}$ be the trace function from E to $\mathbb{F}_q$.

Then χ_1 and μ_1 are related by the identity

$$\chi_1(Tr_{E/\mathbb{F}_q}(\alpha)) = \mu_1(\alpha) \quad \forall\, \alpha \in E$$

Since the trace function satisfies transitivity property we have $\mathrm{Tr}_E = \mathrm{Tr}_{\mathbb{F}_q} \circ \mathrm{Tr}_{E/\mathbb{F}_q}$. Hence for all $\alpha \in E$, $\mu_1(\alpha) = \chi_1(\mathrm{Tr}_{E/\mathbb{F}_q}(\alpha))$.

Similarly if ψ_1 and λ_1 are principle multiplicative characters of $\mathbb{F}_q$ and E respectively then for all $\alpha \in E^*$,

$$\lambda_1(\alpha) = \psi_1(\mathrm{N}_{E/\mathbb{F}_q}(\alpha))$$

where $\mathrm{N}_{E/\mathbb{F}_q}$ is the norm function from E to $\mathbb{F}_q$.

In the opposite way for any additive character χ of $\mathbb{F}_q$ or multiplicative character ψ of $\mathbb{F}_q$ we get lift the characters to an additive and multiplicative character of E respectively by defining $\chi \circ \mathrm{Tr}_{E/\mathbb{F}_q}$ and $\psi \circ \mathrm{N}_{E/\mathbb{F}_q}$ respectively. We denote these by $\chi^{(r)} := \chi \circ \mathrm{Tr}_{E/\mathbb{F}_q}$ and $\psi^{(r)} := \psi \circ \mathrm{N}_{E/\mathbb{F}_q}$ where $r = [E : \mathbb{F}_q]$.

In the following sections we will be working with hybrid character sums which are formed by taking product of multiple characters of both kind. Hence we will be using χ, τ to denote additive characters and ψ, λ to denote multiplicative characters.

1.3 Gaussian Sums

So the first hybrid sum we will be looking at is the Gaussian sum. This is a sum of product of a multiplicative character and an additive character over all elements of the field $\mathbb{F}_q$. So it is defined as follows.

Definition 1.3.1: Gaussian Sum

Let ψ be a multiplicative character and χ be an additive character of $\mathbb{F}_q$. Then the Gaussian Sum *associated to ψ and χ is defined as*

$$G(\psi, \chi) = \sum_{\alpha \in \mathbb{F}_q^*} \psi(\alpha) \cdot \chi(\alpha)$$

Hence the absolute value of the Gaussian sum is at most $q-1$. But later we will see that the Gaussian sum in most cases is much smaller, $\sqrt{q}$ to be precise.

Theorem 1.3.1

Let ψ be a multiplicative character and χ be an additive character of $\mathbb{F}_q$. Then

$$G(\psi,\chi) = \begin{cases} q-1 & \text{if } \psi = \psi_0, \chi = \chi_0 \\ -1 & \text{if } \psi = \psi_0, \chi \neq \chi_0 \\ 0 & \text{if } \psi \neq \psi_0, \chi = \chi_0 \end{cases} \tag{1.4}$$

If ψ is non-trivial multiplicative character and χ is non-trivial additive character then $|G(\psi,\chi)| = \sqrt{q}$.

Proof: If $\psi = \psi_0$ and $\chi = \chi_0$ then for all $\alpha \in \mathbb{F}_q^*$ we have $\psi(\alpha) = 1$ and $\chi(\alpha) = 1$. Hence $G(\psi_0, \chi_0) = q-1$.

Now suppose $\psi = \psi_0$ and χ is non-trivial additive character. Then by Theorem 1.2.2(i)

$$G(\psi_0,\chi) = \sum_{\alpha\in\mathbb{F}_q^*} \chi(\alpha) = \sum_{\alpha\in\mathbb{F}_q} \chi(\alpha) - \chi(0) = -1$$

Let ψ be a non-trivial multiplicative character and $\chi = \chi_0$. Then for all $\alpha \in \mathbb{F}_q^*$ we have $\chi(\alpha) = 1$. Hence by Theorem 1.2.5(i)

$$G(\psi,\chi_0) = \sum_{\alpha\in\mathbb{F}_q^*} \psi(\alpha) = 0$$

So now assume both ψ and χ are non-trivial multiplicative and additive characters of $\mathbb{F}_q$ respectively. Then we have

$$\begin{aligned}
|G(\psi,\chi)|^2 &= \overline{G(\psi,\chi)} \cdot G(\psi,\chi) \\
&= \left(\sum_{\alpha\in\mathbb{F}_q^*} \overline{\psi(\alpha)} \cdot \overline{\chi(\alpha)}\right) \cdot \left(\sum_{\beta\in\mathbb{F}_q^*} \psi(\beta)\cdot\chi(\beta)\right) \\
&= \sum_{\alpha\in\mathbb{F}_q^*}\sum_{\beta\in\mathbb{F}_q^*} \overline{\psi(\alpha)} \cdot \overline{\chi(\alpha)} \cdot \psi(\beta)\cdot\chi(\beta) \\
&= \sum_{\alpha\in\mathbb{F}_q^*}\sum_{\beta\in\mathbb{F}_q^*} \psi(\beta\cdot\alpha^{-1})\cdot\chi(\beta-\alpha) \\
&= \sum_{\alpha\in\mathbb{F}_q^*}\sum_{\gamma\in\mathbb{F}_q^*} \psi(\gamma)\cdot\chi(\alpha(\gamma-1)) && [\text{Take } \gamma = \beta\cdot\alpha^{-1}] \\
&= \sum_{\gamma\in\mathbb{F}_q^*} \psi(\gamma)\cdot\left(\sum_{\alpha\in\mathbb{F}_q^*}\chi(\alpha(\gamma-1))\right) \\
&= \sum_{\gamma\in\mathbb{F}_q^*} \psi(\gamma)\cdot\left(\sum_{\alpha\in\mathbb{F}_q}\chi(\alpha(\gamma-1))\right) - \chi(0)\sum_{\gamma\in\mathbb{F}_q^*}\psi(\gamma) \\
&= \sum_{\gamma\in\mathbb{F}_q^*} \psi(\gamma)\cdot\left(\sum_{\alpha\in\mathbb{F}_q}\chi(\alpha(\gamma-1))\right) && [\textit{Theorem } 1.2.5(i)]
\end{aligned}$$

Hence by Theorem 1.2.2(i)

$$\sum_{\alpha\in\mathbb{F}_q}\chi(\alpha(\gamma-1)) = \begin{cases} 0 & \text{if } \gamma \neq 1 \\ q & \text{if } \gamma = 1 \end{cases}$$

Therefore $|G(\psi,\chi)|^2 = \psi(1)q = q$ and so we have the required result. ■

Now under various transformations of the additive or multiplicative characters Gaussian sums gives some useful identities. We will see some of these below.

Lemma 1.3.2

Gaussian sums for the finite field $\mathbb{F}_q$ satisfy the following identities. For any $a \in \mathbb{F}_q$ let χ_a be the additive character. Let ψ be any multiplicative character and χ be any additive character of $\mathbb{F}_q$. Then we have the following identities

(i) $G(\psi, \chi_{ab}) = \overline{\psi(a)} \cdot G(\psi, \chi_b)$ *for any* $a \in \mathbb{F}_q^*$ *and* $b \in \mathbb{F}_q$.

(ii) $G(\psi, \overline{\chi}) = \psi(-1) \cdot G(\psi, \chi)$.

(iii) $G(\overline{\psi}, \chi) = \psi(-1)\overline{G(\psi, \chi)}$.

(iv) $G(\psi, \chi) \cdot G(\overline{\psi}, \chi) = \psi(-1)q$ *if* ψ *and* χ *are non-trivial multiplicative and additive characters of* $\mathbb{F}_q$ *respectively.*

(v) $G(\psi^p, \chi_b) = G(\psi, \chi_{\sigma(b)})$ *for any* $b \in \mathbb{F}_q$ *where* $p = \mathrm{char}(\mathbb{F}_q)$ *and* $\sigma(X) = X^p$ *be the Frobenius automorphism of* $\mathbb{F}_q$.

Proof:

(i) For any $\gamma \in \mathbb{F}_q$ we have $\chi_{ab}(\gamma) = \chi_1(ab \cdot \gamma) = \chi_b(a \cdot \gamma)$. Hence

$$G(\psi, \chi_{ab}) = \sum_{\gamma \in \mathbb{F}_q^*} \psi(\gamma) \cdot \chi_{ab}(\gamma) = \sum_{\gamma \in \mathbb{F}_q^*} \psi(\gamma) \cdot \chi_b(a \cdot \gamma)$$

Now take $d = a \cdot \gamma$. Then $gm = a^{-1} \cdot d$. Hence

$$G(\psi, \chi_{ab}) = \sum_{d \in \mathbb{F}_q^*} \psi(a^{-1} \cdot d) \cdot \chi_b(d) = \psi(a^{-1}) \cdot \sum_{d \in \mathbb{F}_q^*} \psi(d) \cdot \chi_b(d) = \overline{\psi(a)} \cdot G(\psi, \chi_b)$$

Hence we have the first identity.

(ii) Now we have $\chi = \chi_b$ for some suitable $b \in \mathbb{F}_q$. Therefore $\overline{\chi} = \chi_{(-1)\cdot b}$. Hence by the first identity we have $G(\psi, \overline{\chi}) = \overline{\psi(-1)} \cdot G(\psi, \chi_b) = \psi(-1) \cdot G(\psi, \chi)$. The last equality holds because $\psi(-1) = \pm 1$.

(iii) Now again we have $\overline{\chi} = \chi_b$ for some suitable $b \in \mathbb{F}_q$. Hence $\chi = \chi_{(-1)\cdot b}$. Again by using the first identity we have

$$G(\overline{\psi}, \chi) = G\left(\overline{\psi}, \chi_{(-1)\cdot b}\right) = \overline{\psi(-1)} \cdot G(\overline{\psi}, \chi_b) = \psi(-1) \cdot G(\overline{\psi}, \overline{\chi})$$

Now we definition of Gaussian sums we have $G(\overline{\psi}, \overline{\chi}) = \overline{G(\psi, \chi)}$. Hence we have the required identity.

(iv) Now using part (ii) and (iii) we have

$$G(\psi, \chi) \cdot G(\overline{\psi}, \chi) = G(\psi, \chi) \cdot \psi(-1)\overline{G(\psi, \chi)} = \psi(-1) \cdot |G(\psi, \chi)|^2$$

Since ψ and χ are non-trivial multiplicative and additive characters of $\mathbb{F}_q$ respectively by Theorem 1.3.1 we have $|G(\psi, \chi)|^2 = q$. Hence we have the required identity.

(v) If Tr is the absolute trace function from $\mathbb{F}_q$ to $\mathbb{F}_p$ then for any $\alpha \in \mathbb{F}_q$ we have $\mathrm{Tr}(\alpha) = \mathrm{Tr}(\alpha^p)$. Hence $\chi_1(\alpha) = \chi_1(\alpha^p)$. Hence for any $\alpha \in \mathbb{F}_q$, we get $\chi_b(\alpha) = \chi_1(b \cdot \alpha) = \chi_1(b^p \cdot \alpha^p) = \chi_{\sigma(b)}(\alpha^p)$. Therefore

$$G(\psi^p, \chi_b) = \sum_{\alpha \in \mathbb{F}_q^*} \psi^p(\alpha) \cdot \chi_b(\alpha) = \sum_{\alpha \in \mathbb{F}_q^*} \psi(\alpha^p)\chi_{\sigma(b)}(\alpha^p) = G(\psi, \chi_{\sigma(b)})$$

Therefore we have the required result.

■

Now we can use Gaussian Sums in a variety of context. For any multiplicative character ψ we can express the value of ψ at any nonzero $\alpha \in \mathbb{F}_q$ using Gaussian sums and additive characters. And similarly for any additive character χ of $\mathbb{F}_q$ we can express the value of χ at any nonzero $\alpha \in \mathbb{F}_q$ using Gaussian sums and multiplicative characters.

Now by Theorem 1.2.2(iii) for any $\alpha \in \mathbb{F}_q^*$ we have

$$\begin{aligned}
\psi(\alpha) &= \frac{1}{q} \sum_{\beta \in \mathbb{F}_q^*} \psi(\beta) \sum_{c \in \mathbb{F}_q} \chi_c(\alpha) \overline{\chi_c(\beta)} \\
&= \frac{1}{q} \sum_{c \in \mathbb{F}_q} \chi_c(\alpha) \sum_{\beta \in \mathbb{F}_q^*} \psi(\beta) \cdot \overline{\chi}_c(\beta) \\
&= \frac{1}{q} \sum_{c \in \mathbb{F}_q} \chi_c(\alpha) \cdot G(\psi, \overline{\chi}_c) \\
&= \frac{1}{q} \sum_{\chi \in \boldsymbol{X}_q} G(\psi, \overline{\chi}) \chi(\alpha)
\end{aligned}$$

This may be thought of as the Fourier expansion of ψ in terms of the additive characters of $\mathbb{F}_q$, with Gaussian sums appearing as Fourier coefficients.

We can do similar thing for multiplicative characters of $\mathbb{F}_q$. By using Theorem 1.2.5(iii) we get

$$\chi(\alpha) = \frac{1}{q-1} \sum_{\psi \in \boldsymbol{M}_q} G(\overline{\psi}, \chi) \chi(\alpha)$$

This can be interpreted as the Fourier expansion of the multiplicative characters of $\mathbb{F}_q$ with the Gaussian sums as Fourier coefficients. Hence we have the observations:

Observation 1.7. *For any multiplicative character ψ and additive character χ of $\mathbb{F}_q$ and any $\alpha \in \mathbb{F}_q^*$ we have*

(i) $\psi(\alpha) = \frac{1}{q} \sum_{\chi \in \boldsymbol{X}_q} G(\psi, \overline{\chi}) \chi(\alpha)$

(ii) $\chi(\alpha) = \frac{1}{q-1} \sum_{\psi \in \boldsymbol{M}_q} G(\overline{\psi}, \chi) \psi(\alpha)$

In the next section we will define another hybrid sum called *Jacobi Sums*. Like Gaussian sum this also has many important applications. In the later sections we will use both Gaussian sums and Jacobi Sums to prove many interesting results

1.4 Jacobi Sums

Chapter 2

Bibliography